Supervisor meeting note | 26 October 2026

Sentinel-1 / Sentinel-2 land-cover classification of Porto

SNAP collocation, Random Forest, and why 100 percent accuracy is not a result

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github.com/aleksandrapacia/PortoProject

1. Aim

Test whether collocated Sentinel-1 SAR and Sentinel-2 optical data, processed in SNAP and classified in Python, can separate four land-cover classes over Porto and the Douro valley — water, vegetation, urban, bare soil — and treat accuracy numbers as something to audit, not to advertise.

2. Method (what was actually run)

- SNAP: collocation of Sentinel-1 GRD (sigma0 VV, VH) with Sentinel-2; MNDWI and NDVI as auxiliary layers.
- Labels (v1): pixel thresholds on MNDWI, NDVI and VH. These are not independent ground truth.
- Classifier: Random Forest on raw bands only (B2, B3, B8, B11, sigma0 VV, sigma0 VH). Indices were removed from the feature vector after a 100 percent score showed leakage.
- Validation: stratified random split and spatial block cross-validation (GroupKFold on a 10×10 grid).

3. Results

A random split produced essentially perfect scores (water and vegetation F1 = 1.00; urban about 0.97; overall accuracy about 1.00). Spatial-block CV stayed near 0.99. That is still too high to believe as map accuracy: the labels remain threshold rules on the same image.

What is usable: the classified map is spatially coherent (ocean and Douro as water; built-up as fragmented red; vegetation dominant inland). Feature importance shows optical structure as the main driver, with VH backscatter contributing to the urban class — radar is not redundant.

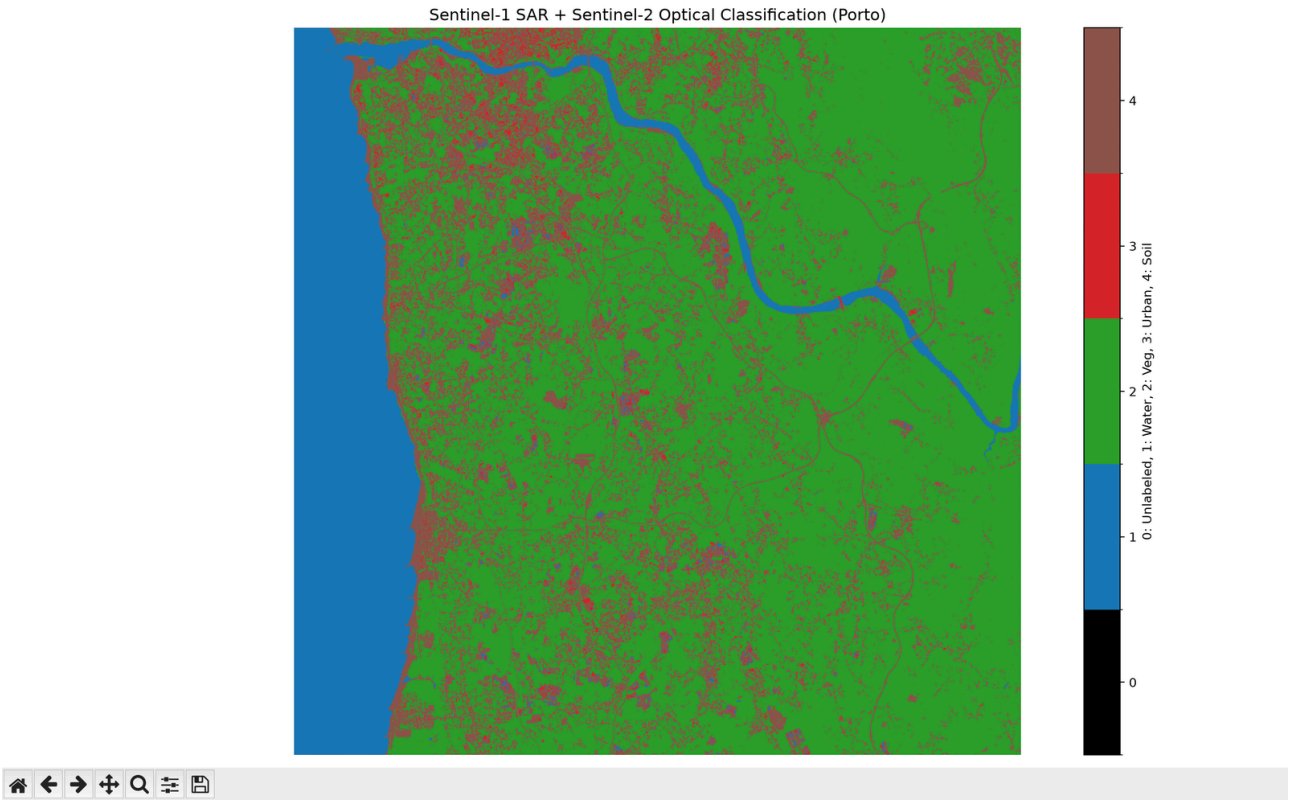


Figure 1. Random Forest map from SNAP-collocated S1+S2. Blue: water; green: vegetation; red: urban; brown: bare soil.

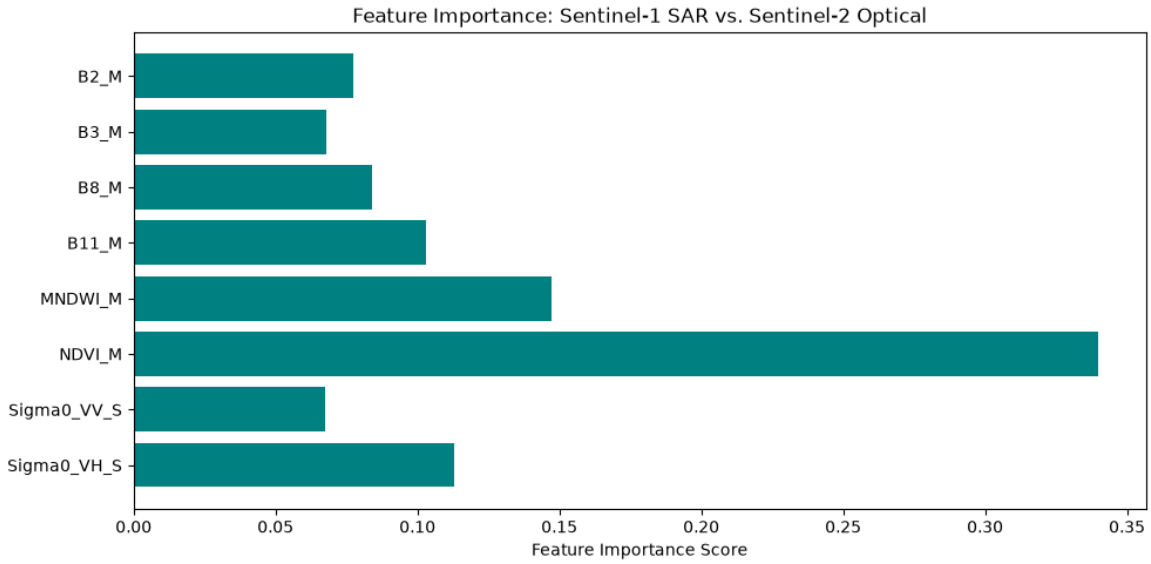


Figure 2. Feature importance (diagnostic run with indices). NDVI dominates; sigma0 VH is the strongest radar contributor.

4. Limitation (the point of this meeting)

The 100 percent figure was a data leak: labels derived from NDVI/MNDWI, then a tree model reproducing that formula, plus neighbouring pixels in both train and test (spatial autocorrelation). Removing indices from training and using block CV is necessary but not sufficient. Independent polygons (hand ROIs, or CORINE / ESA WorldCover) are required before any accuracy is reported as a scientific result.

5. Proposed next steps

- Independent test labels: SNAP/QGIS ROIs versus CORINE or ESA WorldCover.
- Add DEM derivatives (slope, aspect) — Douro valley relief.
- Same spatial CV, second classifier (SVM or XGBoost) for a method comparison, not a

leaderboard.

Question for the supervisor: which of these is the right next increment at Erasmus scale?

I would be grateful for comments on this SNAP–Python workflow, and — if you are willing to guide the next step — for a short recommendation letter in due course. I can provide a one-paragraph description of the method and of my own contribution.